

Environmental cost of AI: we needed real numbers, so here are some

Posted by u/Jygglewag • 0 points • 35 comments

Environmental cost (water, electricity) and pollution (carbon emission) are the biggest things I dislike about generative AI (and crypto mining, but that's a topic for another subreddit).

However, I see people arguing about these impacts without any sort of real data backing their claims. So I'd like to share this short article that measures and compares the impact of genAI to other usual things we do.

the article itself : [<https://journals.calstate.edu/ai-edu/article/view/5448/4391>]
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the article page (for quoting and abstract): <https://journals.calstate.edu/ai-edu/article/view/5448>

Comments

u/Dry_Caterpillar2207 • 1 points • 2025-09-22 16:29 UTC

1 hamburger is equal to 15.5kg CO2e

But youre already a vegan, right? Because you care about the earth so much?

u/TheSinhound • 1 points • 2025-09-19 02:32 UTC

I keep having to say this, but the cost is negligible at best. Global agriculture water usage is around 2 quadrillion gallons of water a year (And we waste about 18% of the food we create with it). EVERY DATA CENTER ON EARTH IN TOTAL consumes about 500 billion gallons (As in, for everything, not just AI tasks). Water use is a fraction of a fraction. Power use is about 1.8% globally for AI tasks. GHG emissions are about 1% of global. It's absolutely an issue on a local scale, but globally it's so low as to be statistically insignificant. AND it's only becoming more efficient. Yes, demand will increase, but even at 10x of current use it's not going to move the needle.

u/Victormorga • 1 points • 2025-09-18 20:05 UTC

"At scale, however, generative AI contributes notably to global energy and water demands."

It's one thing to say the environmental impact is incorrectly calculated or even overstated, but minimizing the effects is as disingenuous as saying it will destroy the world in a year.

u/SlapstickMojo • 4 points • 2025-09-18 18:40 UTC

Let's see... article states:

"A single query to ChatGPT emits approximately 4.3 grams of CO₂ (Patterson et al., 2021)".

Bibliography: Patterson, D., Gonzalez, J., Le, Q. V., Liang, C., Munguia, L. M., Rothchild, D., ... & Dean, J. (2021). Carbon Emissions and Large Neural Network Training. Communications of the ACM, 64(4), 58-67.

Referenced paper here: <https://arxiv.org/pdf/2104.10350>

Right off the bat, the paper cited is 4 years old -- a lifetime in this field.

Quick scan... "4.3" shows up seven times in the article:

1. "Section 4.3"
2. "Google's custom TPU v2 processor runs Transformer (Big) 4.3X faster than P100 GPUs and Evolved Transformer (Medium) 5.2X faster."
3. "Sparse models can have many model parameters while requiring much less computation than dense models. Training GShard-600B used 24 MWh and produced 4.3 net tCO₂e."
4. "Table 4: Model Gshard -600B; Net tCO₂e for Model Training: 4.3"
5. "Section 4.3"
6. "GShard-600B's emissions (Table 4) are 4.3 tCO₂e —3.5 passenger SF-NY round trips—from consuming 24 MWh to train the model that could have 2B users; the amortized per-user CO₂e impact of model training would be less than the CO₂e impact of sending one text message."
7. "Section 4.3"

No results for the word "query".

2 results for the word "queries":

1. "Despite targeting a narrow audience of scientists, COVID explorer served 1000 queries per day at launch."
2. "...our large scale translation models (M4) have already been used to translate billions of queries annually for each mid-to-low resource language..."

Seems like a major problem right off the bat. Either wrong article cited, misread citation, or fabricated. That's the baseline, too, so one should ask why to trust anything else in the article.

u/Kathane37 • 4 points • 2025-09-18 17:22 UTC

« A single query to ChatGPT emits approximately 4.3 grams of CO₂ (Patterson et al., 2021 »

Pure trash article that base it's calculation on extrapolation over gpt-3 architecture (a large dense model)

While gpt-4 was already suspected to be a MoE and that now every single model on the market is a MoE using way less energy since there is way less active parameters

u/2008knight • 1 points • 2025-09-18 15:47 UTC

I want to compare it to Baldur's Gate 3 specifically because when I run SDXL locally on my computer, the fans just spin a little bit faster. But when I start Baldur's Gate, the damned thing sounds like it's about to take off.

u/Vaughn • 1 points • 2025-09-18 19:27 UTC

The fans **should** spin up considerably. If they don't, that sounds like something is bottlenecking SDXL and keeping it from running at a reasonable speed -- most likely low memory.

u/2008knight • 1 points • 2025-09-19 05:43 UTC

I'm just running SDXL on 16 gb of VRAM. I have VRAM to spare.

The issue with Baldur's Gate is probably that it makes my CPU ramp up as well. When running SDXL I don't really need to use much CPU at all.

u/Vaughn • 1 points • 2025-09-19 11:40 UTC

Hmm. Well, if you're on Linux you can run `\`nvidia-smi\`` for a quick check of GPU utilization.

I think windows has nvidia-smi to do something similar. Don't use it.